

Chapter 9

Alkynes

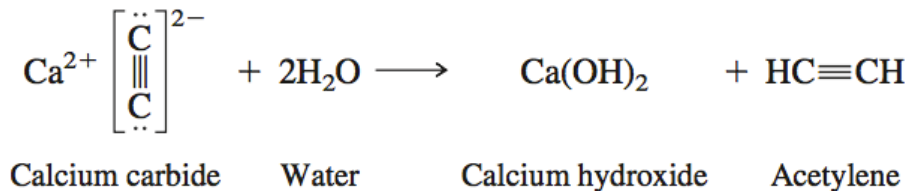
Additional Recommended Problems: Chp9: 8, 17,20,24,25,27,30-32,34,36

CH 9 Learning Objectives

- Correctly name alkynes.
- Describe the structure and properties of alkynes.
- Rank the acidity strength between alkanes, alkenes, and alkynes and be able to describe why sp hybridized carbons are the most acidic.
- Know the products, including regioselectivity and stereochemistry if needed, for the following reactions with alkynes.
 1. Preparation of Alkynes (Table 9.2)
 1. S_N2 Reactions (form C-C bond)
 2. E2 Reactions
 2. Hydrogenation
 - A. Alkyne $\rightarrow$ Alkane
 - B. Alkyne $\rightarrow$ Alkene
 3. Addition of HX
 4. Hydration of Alkynes/Oxymercuration
 5. Hydroboration/Oxidation
 6. Addition of X_2
 7. Ozonolysis
- Use retrosynthetic analysis to solve multi-step synthesis design problems involving alkynes and alkenes as intermediates or final products

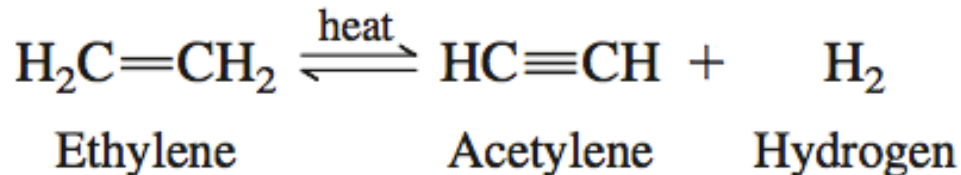
Alkynes

- *Acetylene* (C_2H_2 , *ethyne*) can be produced in large quantities from calcium carbide



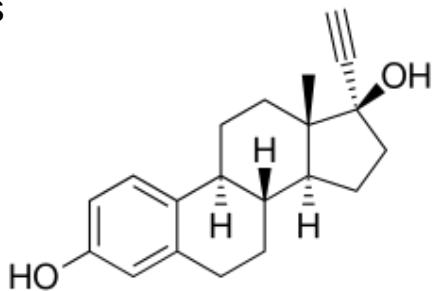
– <https://www.acs.org/content/acs/en/education/whatischemistry/landmarks/calciumcarbideacetylene.html>

- Acetylene can also be produced via *dehydrogenation* of ethene

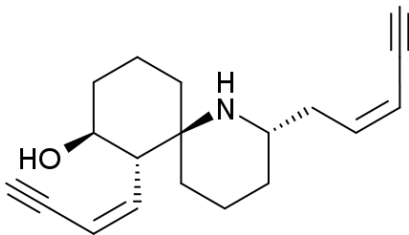


Alkynes

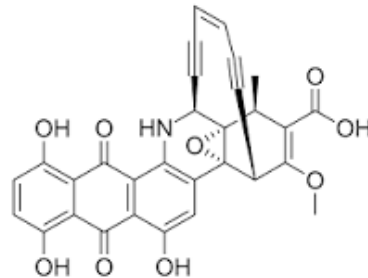
- Ethynylestradiol is an estrogen medication widely used in birth control pills



- Histrionicotoxins are produced on the skin of poison frogs



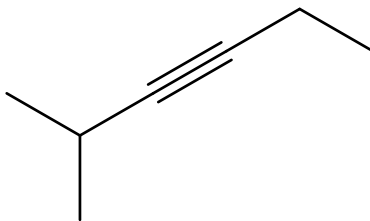
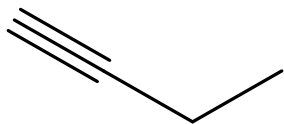
- Dynemicin A is a promising anti-cancer drug



- Acetylene gas burns with very bright and very hot flame
 - used in oxygen-acetylene welding
 - Headlights for early automobiles
 - Lights on helmets for miners

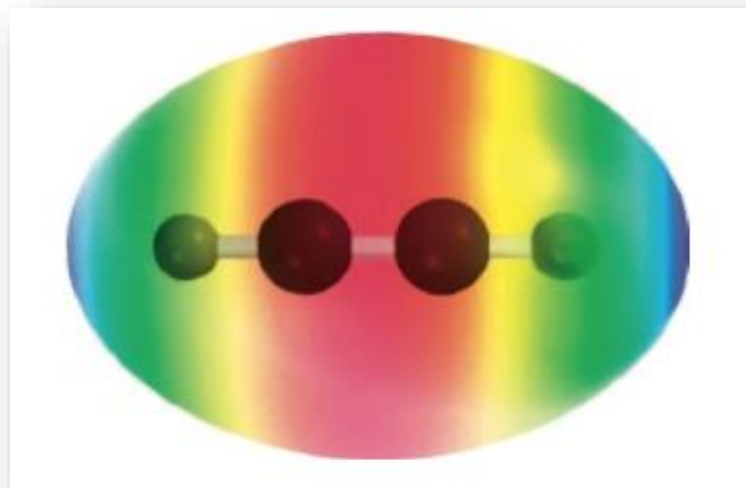
Alkyne Nomenclature

- Triple bond has the same priority as alkene when numbering
 - Replace the parent chain name –ane with –yne



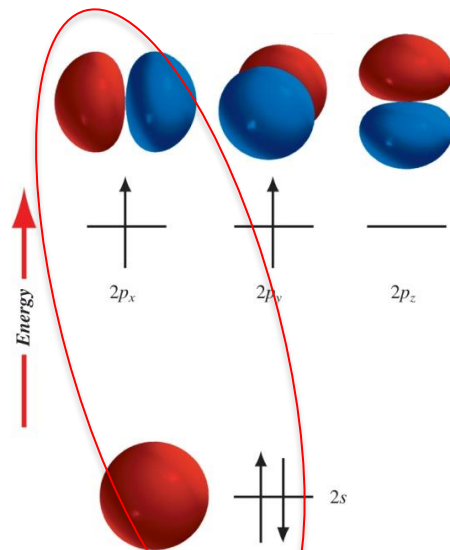
Alkyne Properties

- Physical properties of alkynes are similar to those of alkanes and alkenes of comparable molecular weight
 - Low polarity
 - Low boiling points
 - Low density
 - Low solubility in water
 - (when lacking other polar functional groups)



Structure/Hybridization – sp

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(a) Most stable electron configuration of carbon atom

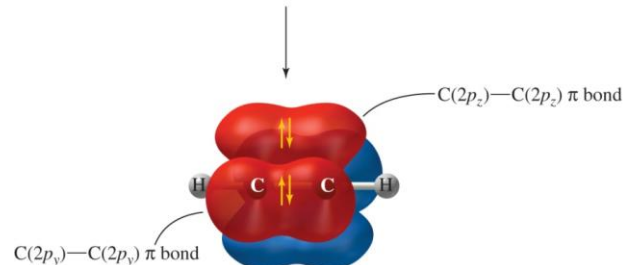
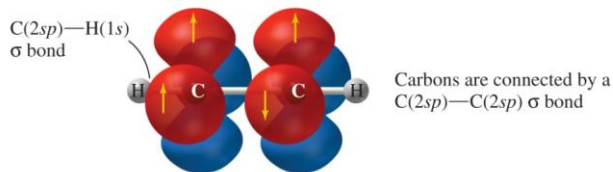
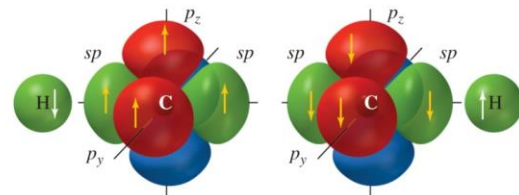
(b) *sp* Hybrid state of carbon

Structure of Ethyne (C_2H_2)



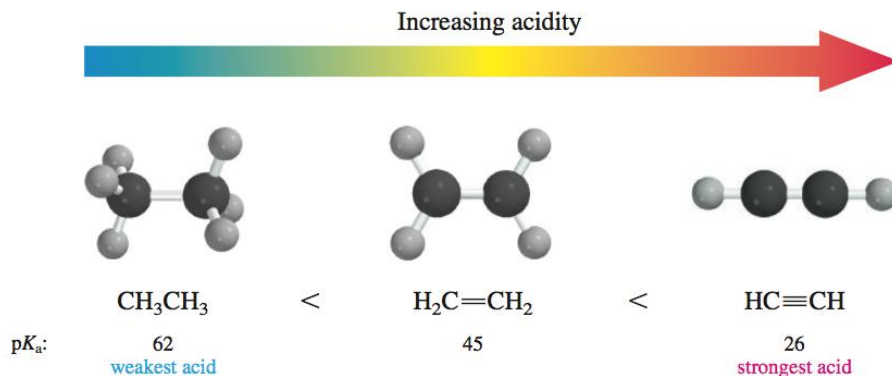
- **Molecular Geometry: linear**
- **bond angles: 180°**
- **bond distances: C—H = 106 pm**
C $\equiv$ C = 120 pm

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Acidity of Alkynes/Alkenes/Alkanes

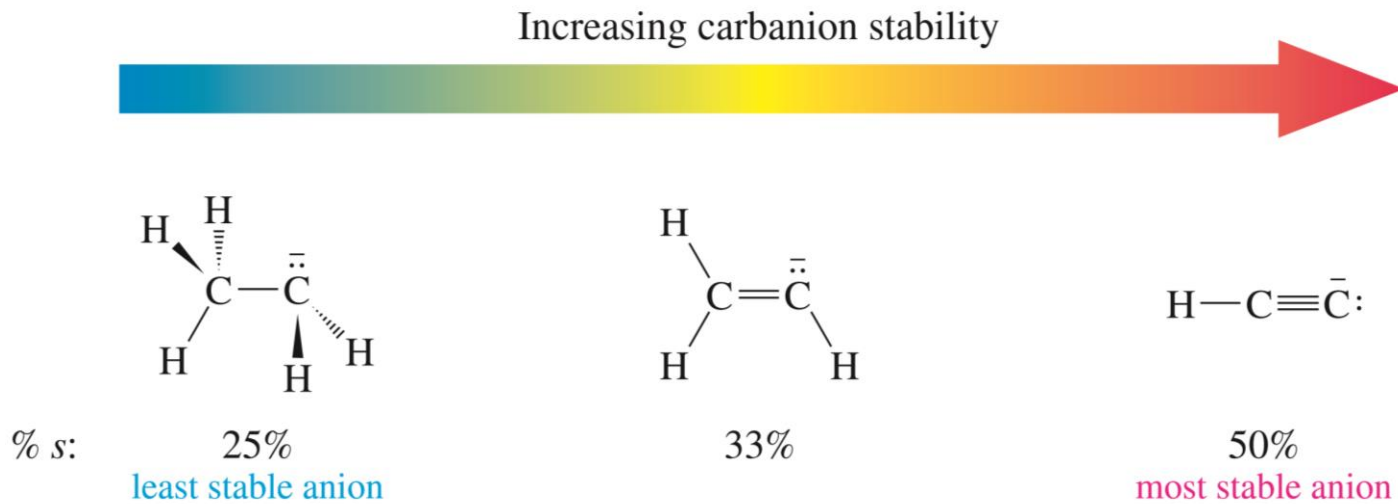
- Relative to alkanes and alkenes, terminal alkynes are remarkably acidic.



- Deprotonation of ethyne leaves a lone pair in a hybrid orbital with high s character. (higher s character can better stabilize the negative charge)

Carbanion Stability

- High s character means lone pair can get closer to the positively charged nucleus



Acid/Base Review

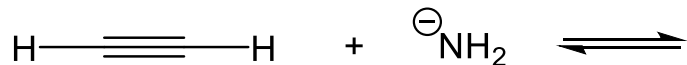
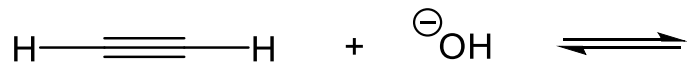
Short pK_a table

Functional group	Example	Stronger acid	pK _a	conjugate base	Weaker base
Hydroiodic acid	$\text{H}-\text{I}:$		-9	$:\ddot{\text{I}}:^{\ominus}$	
Carboxylic acids	$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{OH}$		5	$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{O}^{\ominus}$	
Protonated amines	$\text{NH}_4^+ :\ddot{\text{Cl}}:^{\ominus}$		10	$:\text{NH}_3$	
Thiol	$\text{R}-\ddot{\text{S}}-\text{H}$			$\text{R}-\ddot{\text{S}}:^{\ominus}$	
Water	$\text{HO}-\text{H}$		14	$^{\ominus}\text{OH}:$	
Alkyne	$\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{H}$		25	$\text{R}-\text{C}\equiv\text{C}:^{\ominus}$	
Amine	$:\text{NH}_3$		~35	$^{\ominus}\text{NH}_2:$	
Hydrogen	$\text{H}-\text{H}$			$\text{H}^{\ominus}:$	
Alkane	CH_4		~50	$^{\ominus}\text{CH}_3:$	

↓ Weaker acid
↑ Stronger base

Slide 52 CH 1: Strength of Acid Examples

- Complete the following reactions. Given that ethyne as a pKa value of 25, water has a pKa value of 15.7, and ammonia has a pKa value of 36, determine whether the reactants or products are favored for each reaction.



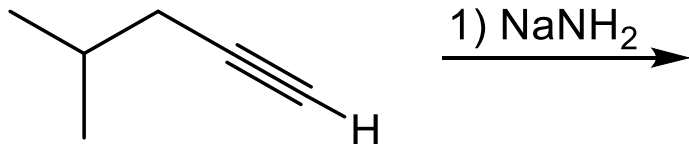
- Which is a more effective base to deprotonate ethyne?

Alkyne Reactions

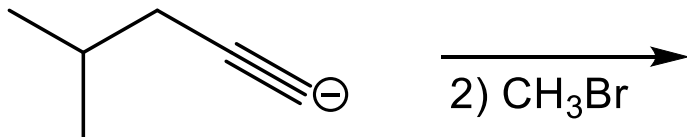
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7. Ozonolysis

1. Preparation of Alkynes (S_N2)

- Alkylation of terminal alkynes – replaces an $-H$ or other functional group with an alkyl group

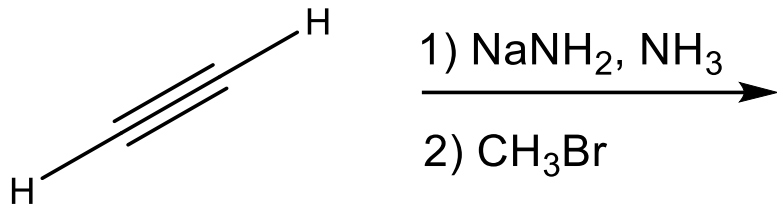
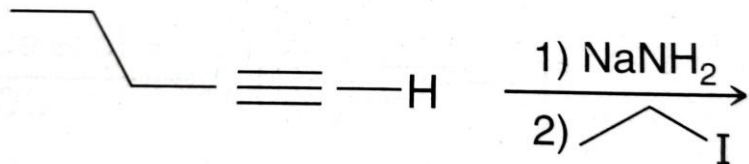


- Acetylide anion is a strong nucleophile (S_N2 for $\text{CH}_3\text{-X}$ or 1°R-X)



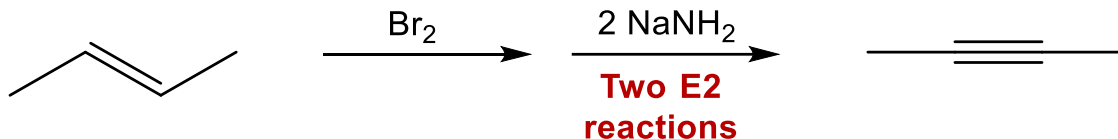
1. Preparation of Alkynes (S_N2) Example

- Predict the product for each of the following reactions.



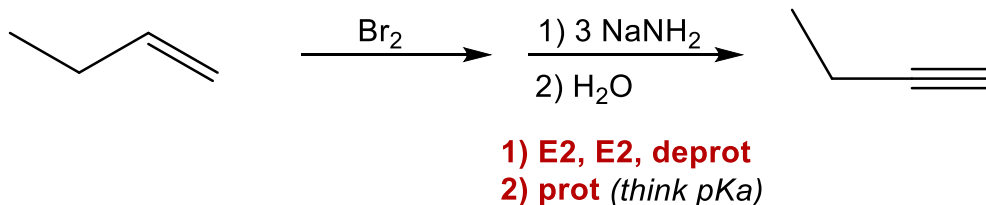
1. Preparation of Internal Alkynes (E2)

- Alkynes can be synthesized from alkenes.
 - Addition of X_2 followed by successive E2 eliminations
 - Internal alkynes: 2 equivalents of NaNH_2



1. Preparation of Terminal Alkynes (E2)

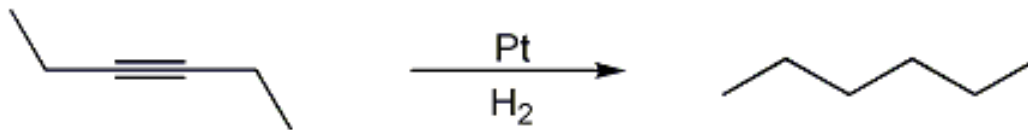
- Alkynes can be synthesized from alkenes.
 - Addition of X_2 followed by successive E2 eliminations
 - Terminal alkynes: 3 equivalents of NaNH_2 , then water



2. Hydrogenation (to alkane)

A. Alkyne $\rightarrow$ Alkane

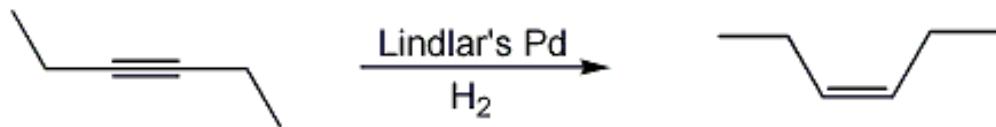
- Addition of H_2 across triple bond, generates alkane product (*reduction of $\text{C}\equiv\text{C}$*)
- Mechanism – heterogenous catalysis, H_2 and alkyne adsorbed to surface of metal (Pt common)



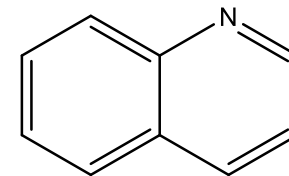
2. Hydrogenation (to alkene)

B. Alkyne $\rightarrow$ Alkene

- Generate **cis**-alkenes with Lindlar's catalyst (poisoned Pd) and H_2



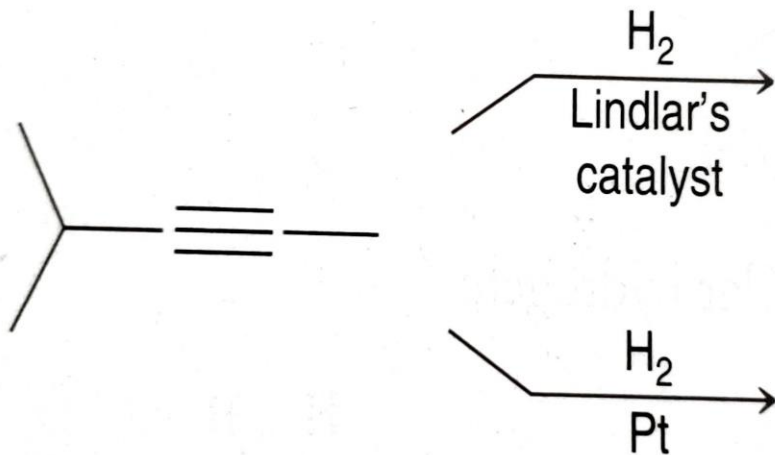
- Pd on calcium carbonate has been “poisoned” with added lead acetate and quinoline to make it not as effective.



quinoline

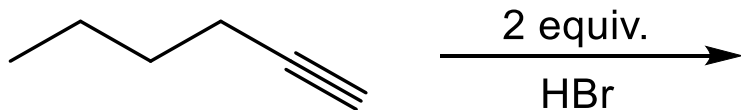
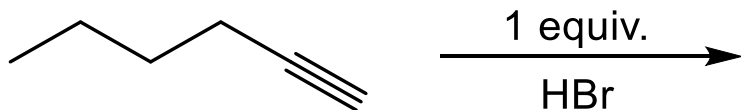
Hydrogenation Example

- Predict the major products below.



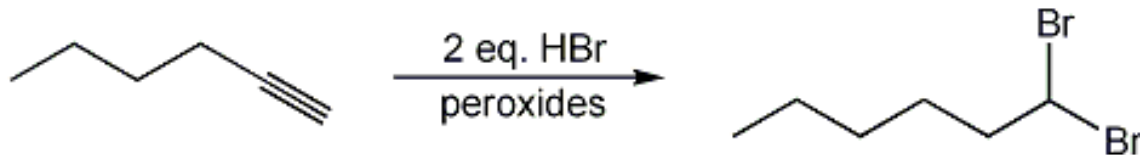
3. Addition of HX to Alkynes

- 1 eq. gives alkene, 2 eq. gives *geminal* dihalide
- Follows Markovnikov rules



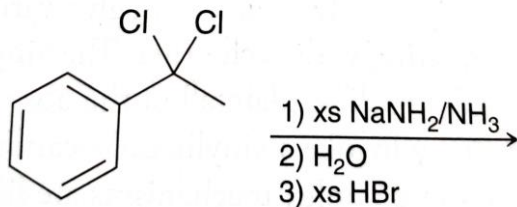
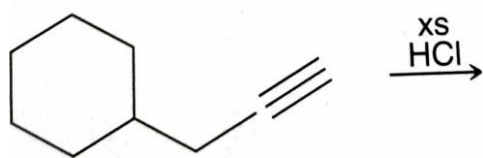
3. Addition of HX to Alkynes (1)

- HBr with peroxides (anti-Markovnikov)



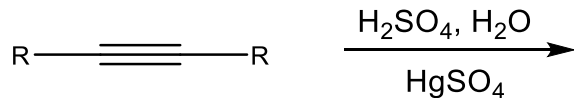
Examples

- Predict the major product for each of the following.

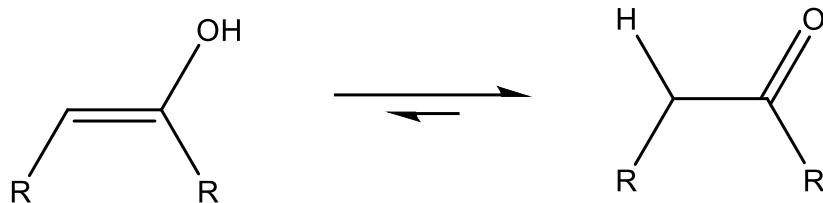


4. Hydration of Alkynes

- Acid catalyzed reaction
 - Mercury sulfate is a catalyst that makes the reaction go faster and in higher yield



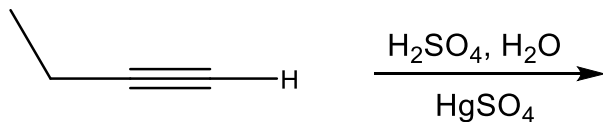
Enol-Keto Tautomerism



- Enols are isomers of ketones and exist in equilibrium with them
 - These are called tautomers
- Keto-Enol equilibration is fast in acidic or basic solutions
- Ketones are usually more stable than enols, therefore the ketone isomers are favored at equilibrium

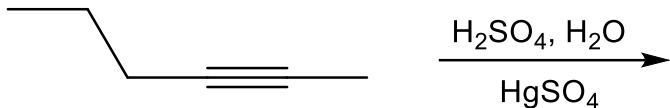
4. Hydration of Alkynes – Terminal Alkynes

- Follow's Markovnikov's Rule

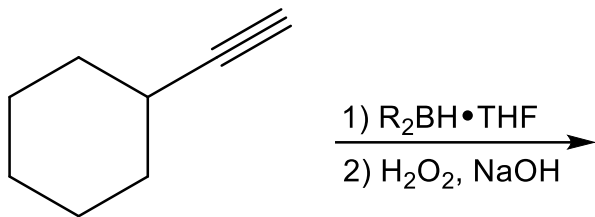


4. Hydration of Alkynes – Internal Unsymmetric Alkynes

- Because protonation at either end gives carbocations of similar stability, a mixture of two constitutional isomers is usually formed

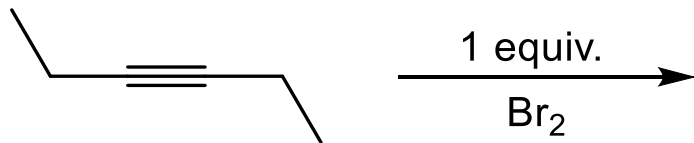


5. Hydroboration/Oxidation



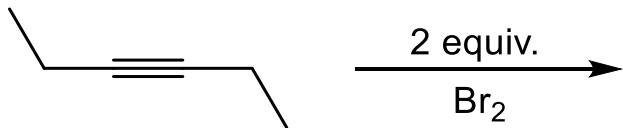
6. Addition of X₂

- 1 equivalent:



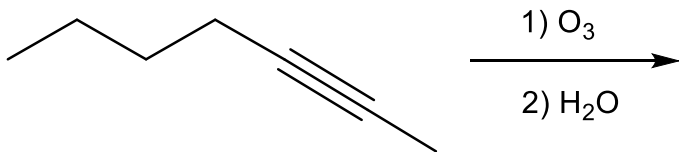
6. Addition of X_2 (1)

- 2 equivalents:

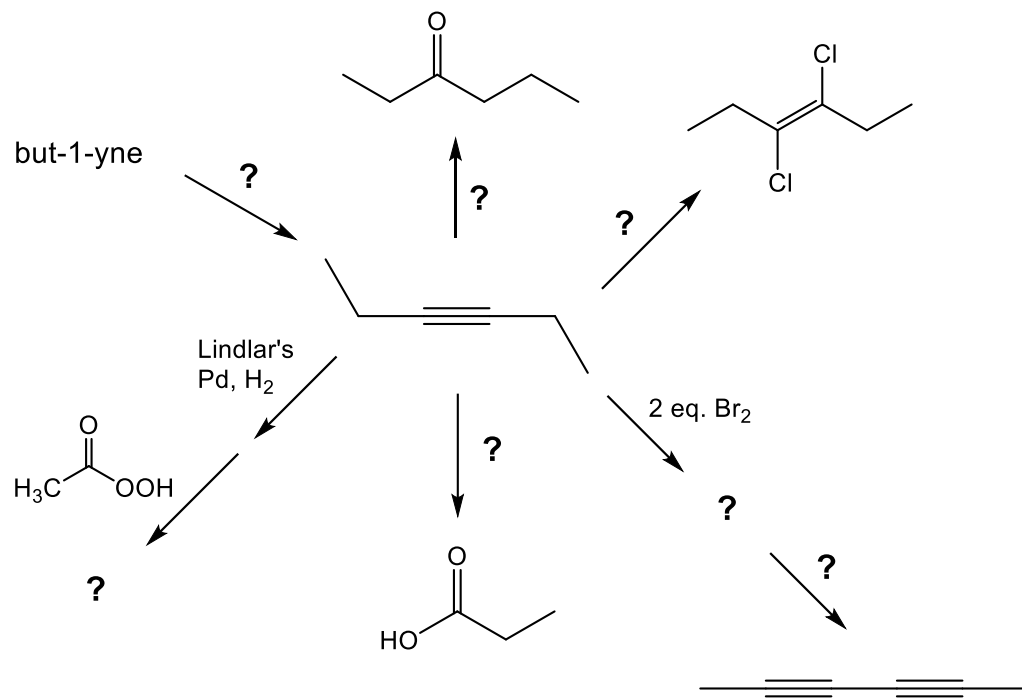


7. Ozonolysis

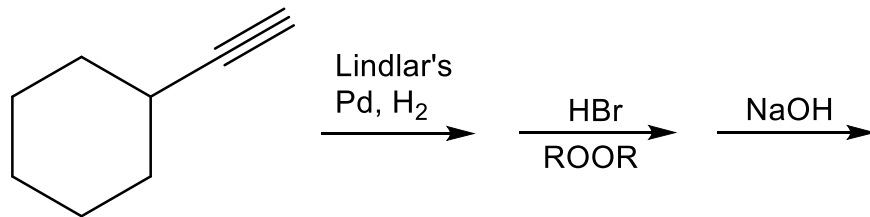
- Reaction of an alkyne with ozone followed by water leads to the production of carboxylic acids.



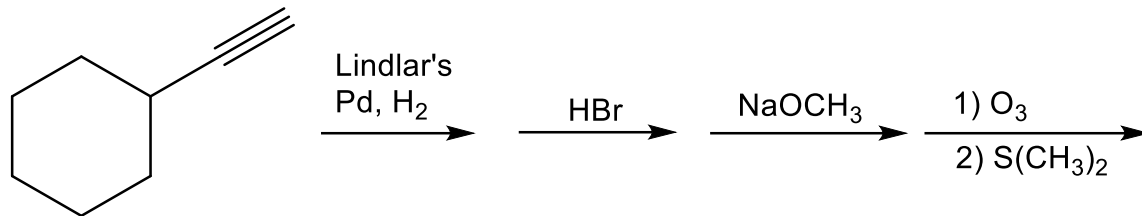
Draw the products or reagents that are missing for the following reactions



Draw the products for the following reactions



Draw the products for the following reactions



Devise a Method for the Following

